Topics Covered:

- Agents
- Tree Search
- Graph Search

Question 1: Agents

a) Types of Agents

Explain the four types of agents:

1) Simple-Reflex Agent

2) Model-Based Agent 3) Goal-Based Agent 4) Utility-Based Agent.

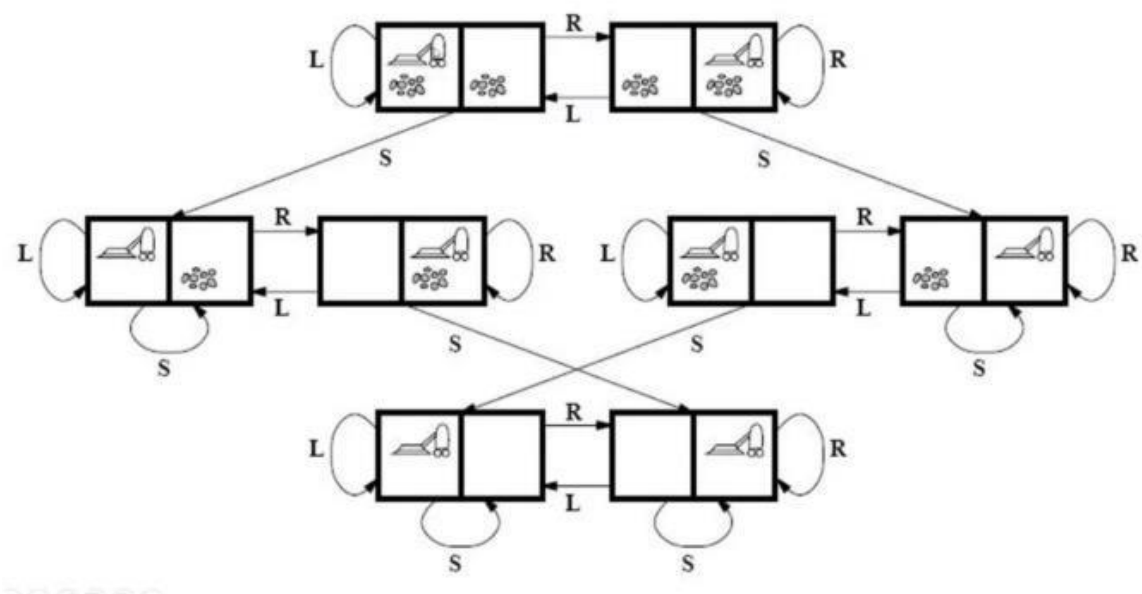
Provide a brief description of each type and discuss their characteristics.

b) Vaccuum Cleaner Agent

Consider the classic vacuum cleaner world problem where a vacuum cleaner agent operates in a 2x2 grid environment. The grid can be in one of two states: dirty or clean. The agent can move left, right, up, or down and can perform an action to either suck dirt or move to an adjacent square.

Implement a simple reflex agent of a vacuum cleaner. For coding, you are not allowed to use any library (agents.py). Although, you can use other python libraries for help.

The states and actions of the agent are provided below:



Actions:

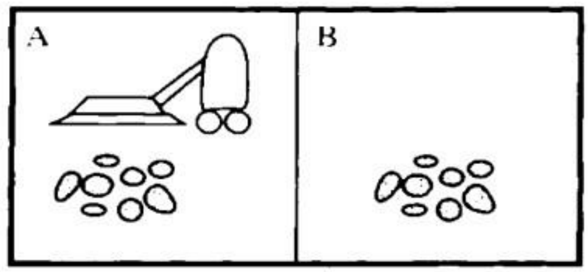
- Move Right [R]
- Move Left [L]
- Suck Dirt [S]

Start State: Any state from Room [A] or [B].

End State: Both rooms are cleaned.

Percept Sequence: Use the following percept sequence.

Percept sequence	Action
[A, Clean]	Right
[A, Dirty]	Suck
[B, Clean]	Left
[B, Dirty]	Suck



c) Identifying Agents

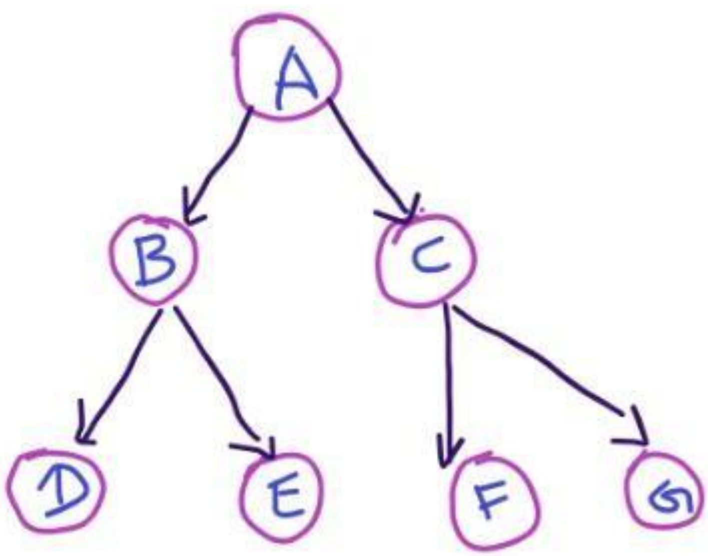
For the following scenarios, discuss which type of agents (reflex, model-based, goal-based, utility-based) would be most suitable depending upon the nature of the environment.

1) **Smart Home Systems:** Consider a smart home system equipped with various IoT devices such as smart thermostats, lights, and security cameras. The system needs to monitor and adjust environmental conditions, provide security, and optimize energy usage.
2) **Autonomous Delivery Robots:** Imagine a scenario where an ecommerce company wants to deploy autonomous delivery robots to deliver packages within a city. These robots need to navigate through city streets, avoid obstacles, and deliver packages to the correct addresses efficiently.
3) **Medical Diagnosis Systems:** In the field of healthcare, medical diagnosis systems aim to assist healthcare professionals in diagnosing diseases and recommending treatment plans based on patient symptoms and medical history.

4) **Autonomous Vehicles:** Autonomous vehicles, such as self-driving cars, aim to navigate roads safely and efficiently without human intervention.
5) **Financial Trading Systems:** Financial trading systems use algorithms to analyze market data and execute trades on behalf of investors.

Question 2: Tree Search Algorithms

Consider the following tree:



a) Uninformed Search

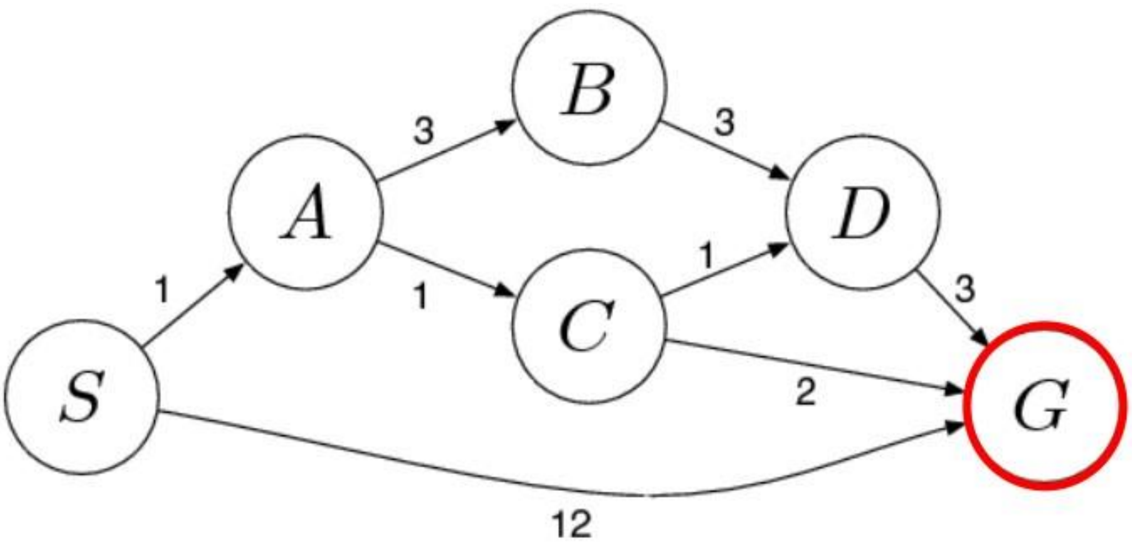
Perform the following searching techniques on the given tree:

1) Breadth-First Search (BFS)
2) Depth-First Search (DFS)
3) Iterative Deepening Search (IDS)
4) Depth-Limited Search (DLS)

You have to find the shortest path from node A to node G (goal node). Show the order in which the nodes are visited during each algorithm.

Question 3: Graph Search Algorithms

Consider the following graph:



a) Uniform Cost Search

In graph, S is the starting node and G is the goal node. Perform Uniform Cost Search (UCS) on the graph. Show the order in which nodes are visited and the cost associated with each path during the search.